

Memory Utility Networks

An Empirical Investigation of Utility-Based Memory Retrieval

Technical Report — Revised Edition

Phases 1 through 13C.9 · Research Concluded

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Research Status

Complete — Negative Result Documented — Artifacts Archived

Research Overview

Figure 1. Research Timeline: Thirteen Experimental Phases



Figure 1. Research Timeline

Research timeline spanning thirteen experimental phases across four macro-stages: Foundation (Phases 1–11), Validation and Audit (Phase 12), MUN v2 Label-Free Investigation (Phase 13B–13C), and Final Controlled Validation (Phase 13C.9). The programme concluded with a documented positive result for MUN v1 and a documented negative result for MUN v2.

Abstract

Memory retrieval is a foundational challenge in the design of intelligent systems. Whether instantiated as few-shot in-context learning, retrieval-augmented generation, or long-horizon agent memory management, the capacity to surface the right memory at the right moment determines both the quality and efficiency of downstream reasoning. Existing retrieval strategies — cosine similarity, recency ordering, access frequency, and their variants — are principled but fundamentally reactive heuristics: they score memories based on historical patterns rather than prospective utility. They answer the question of *what a memory has been*, not *what it will be worth when retrieved in context*.

This report documents a complete, multi-phase empirical investigation of utility-based memory retrieval under the **Memory Utility Networks (MUN)** framework. The central hypothesis holds that the future usefulness of a memory, given a query context, can be predicted directly by a learned function — outperforming heuristic retrieval strategies. We designed, implemented, validated, and iterated upon two generations of MUN models applied to the few-shot sentiment classification task on the Stanford Sentiment Treebank v2 (SST-2) benchmark.

MUN v1, a supervised neural utility scorer employing label-boosting, temporal decay, contrastive loss, context-aware attention over the memory pool, and curriculum-based training, achieved **Recall@5 = 0.829** and **NDCG@5 = 0.914**, averaged across five independent random seeds. These results substantially and significantly outperform seven standard retrieval baselines, with the comparison against the LRU baseline yielding $t(4) = 45.56$, $p < 10^{-10}$, Cohen's $d \approx 28.8$. A seven-question validation audit (Phase 12D) confirmed the absence of data leakage, label leakage, or implementation error. Ablation studies (Phase 12E) revealed that approximately 34 percentage points of MUN v1's accuracy advantage derive from access to query-label information during memory selection — an intentional architectural design choice that nonetheless limits applicability in label-free deployment contexts.

MUN v2, designed to achieve competitive retrieval performance in label-free settings, was developed across Phases 13B through 13C.9. Seventeen candidate label-free features were evaluated on 2,000 (query, memory, utility) triplets. Text overlap (Jaccard similarity, Pearson $r = 0.716$) emerged as the dominant predictor. Semantic embedding features derived from SentenceTransformer (all-MiniLM-L6-v2, 384 dimensions) showed only weak signal (best $r = 0.194$), below the threshold required to justify a neural architecture. A systematic weight optimisation over 128 configurations and a feature ablation across eight subsets were conducted. Despite these efforts, the final controlled validation (Phase 13C.9, $N = 20$, seed fixed at 42) found MUN v2.1 achieving **35% accuracy** — performing below the similarity baseline (65%), below recency ordering (60%), and even below random memory selection (40%). The pre-registered decision criterion requiring MUN v2 to exceed the similarity baseline was not satisfied; MUN v2.1 development was formally concluded as a documented negative result.

A significant methodological contribution is a **45-percentage-point reproducibility discrepancy** (Phase 13C.8) that emerged between intermediate evaluation phases, attributable to uncontrolled weight configuration differences amplified by small test set size. This incident was investigated in detail, partially resolved, and documented as a case study in the fragility of small-N empirical evaluation. We conclude that label-aware utility estimation is demonstrably effective in the SST-2 few-shot classification setting; label-free utility estimation via engineered text and information-theoretic features is not sufficient to exceed cosine similarity in this context. Utility-based memory retrieval remains a promising but unresolved research challenge.

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1. Introduction

1.1 Background and Motivation

The management of memory is a pervasive challenge in the design of intelligent systems. For neural language models, memory manifests as the *context window*: a fixed-capacity token budget that must be allocated among prior conversation turns, retrieved documents, tool outputs, and task instructions. For retrieval-augmented systems, it manifests as *document selection*: which passages from a knowledge base should inform the generative model's response. For autonomous agents operating over extended time horizons, it manifests as *episodic memory management*: which prior observations, reasoning steps, and tool interactions are worth preserving versus evicting as the agent accumulates experience.

Across all these settings, the quality of the memory retrieval decision directly gates downstream performance. Numerous empirical studies have demonstrated that the choice of which examples appear in the context window of a few-shot language model can account for performance swings of 10 to 40 percentage points on standard benchmarks — a range that dwarfs the improvements typically claimed by architectural innovations (Brown et al., 2020; Liu et al., 2021). Yet the dominant retrieval paradigms remain unchanged: cosine similarity over dense embeddings, recency-based queuing, or access frequency counting. These strategies are computationally efficient and often effective, but share a fundamental limitation: they are *retrospective*. They measure how similar a memory is to the current query, or how recently or frequently it has been accessed, but they do not estimate whether retrieving that memory will actually improve the system's output.

The Memory Utility Networks project was initiated to investigate whether this gap can be closed — whether a learning algorithm can acquire a forward-looking utility estimator that predicts, given a query context, which memory from a pool will most improve performance on the downstream task if retrieved. This investigation spanned thirteen experimental phases, producing two generations of models, a multi-layer audit protocol, a systematic feature engineering study, and ultimately a documented negative result for the label-free variant.

1.2 Formal Problem Statement

We formalise the memory selection problem as follows. Let q denote a query, and let $M = \{m_1, m_2, \dots, m_n\}$ denote a memory pool where each memory $m_i = (\text{text}_i, \text{label}_i)$ consists of a text string and an associated label. The goal is to select a subset $S \subseteq M$ of size k such that the downstream performance of a task model Θ operating with S as in-context context is maximised:

$$S^* = \operatorname{argmax}_{S \subseteq M, |S|=k} E[\operatorname{Perf}(\Theta, q, S)]$$

The utility of a single memory m_i given query q is then approximately:

$$U(q, m_i) \approx E[\operatorname{Perf}(\Theta, q, S \cup \{m_i\})] - E[\operatorname{Perf}(\Theta, q, S)]$$

The MUN framework aims to learn an approximation $\hat{U}(q, m_i)$ of this utility function from supervision, such that ranking the pool by $\hat{U}$ and retrieving the top- k elements achieves performance close to the oracle selection S^* .

1.3 Research Questions

- **RQ1 (Efficacy):** Can a learned utility scorer $\hat{U}(q, m)$ outperform heuristic retrieval strategies — including cosine similarity, recency, frequency, LRU, FIFO, TF-IDF, and random selection — on a standardised few-shot classification benchmark?
- **RQ2 (Attribution):** What fraction of the utility signal in MUN v1 derives from label information versus semantic text features? Is the label boost an essential or incidental component?
- **RQ3 (Label-Free Feasibility):** Can label-free utility estimation — relying solely on text-based and information-theoretic features — achieve competitive retrieval performance without access to query or memory labels?
- **RQ4 (Feature Importance):** Among candidate label-free features, which carry the strongest predictive signal for utility, and how does embedding-based semantic similarity compare with surface-level text overlap?

1.4 Contributions

- **MUN v1:** A multi-component neural utility scorer combining similarity scoring, temporal decay, contrastive loss, context-aware attention, and curriculum-based training, achieving Recall@5 = 0.829 and NDCG@5 = 0.914 (5 seeds, $p < 10^{-10}$).
- **MUN v2:** A transparent, label-free, feature-weighted utility scoring model derived from a systematic 17-feature correlation analysis on 2,000 labelled triplets.
- **Audit Protocol:** A multi-phase validation framework comprising data leakage checks, label leakage analysis, metric verification, and independent replication studies.
- **Reproducibility Case Study:** A documented 45-percentage-point discrepancy between evaluation phases, investigated and partially resolved, with methodological implications for small-N evaluation in machine learning.
- **Documented Negative Result:** MUN v2.1 achieves 35% accuracy under controlled conditions, failing the pre-registered criterion and providing a rigorous counter-example to naive optimism about label-free utility estimation.

2. Background and Related Work

2.1 Memory Networks and Early Neural Memory Systems

The Memory Networks architecture (Weston et al., 2014) introduced the core abstraction of a persistent, addressable external memory component coupled to a neural inference network. Memory Networks employed soft attention over stored memory slots to retrieve relevant information for question answering, establishing the paradigm of differentiable memory read operations. End-to-End Memory Networks (Sukhbaatar et al., 2015) extended this with fully differentiable memory addressing via multiple-hop reasoning.

The Neural Turing Machine (Graves et al., 2014) equipped a recurrent controller with read and write heads operating on a structured memory matrix. Content-based addressing in the NTM — comparing controller state to memory entries via cosine similarity — is the conceptual antecedent of similarity-based retrieval in modern embedding search systems. The Differentiable Neural Computer (Graves et al., 2016) extended the NTM with dynamic memory allocation via usage tracking and temporal link matrices.

MUN departs from this tradition by *decoupling* the retrieval decision from the model architecture. Rather than learning a memory addressing mechanism as part of the model's forward pass, MUN learns a standalone utility scorer applicable to any memory pool with any downstream model. This modularity enables evaluation against non-neural baselines and facilitates systematic ablation and audit.

2.2 Retrieval-Augmented Generation

Lewis et al. (2020) demonstrated that RAG — combining a Dense Passage Retrieval (DPR) module with a seq2seq generation model — substantially improves open-domain question answering over purely parametric models. REALM (Guu et al., 2020) pre-trained the retrieval module jointly with the language model, propagating gradients from the generation loss back through the retrieval step. This approach is conceptually similar to the utility-based framing of MUN; the key difference is that REALM trains in a large-data regime, while MUN targets the low-resource few-shot setting.

Dense Passage Retrieval (Karpukhin et al., 2020) established the dual-encoder architecture as the standard approach for open-domain retrieval, demonstrating that learned dense representations substantially outperform BM25 sparse retrieval. DPR, however, retrieves based on query-passage similarity and does not model the downstream utility of the retrieved passage for the generation task — the precise gap that MUN targets.

2.3 In-Context Learning and Example Selection

Brown et al. (2020) demonstrated that large language models can perform tasks from a small number of in-context examples without parameter updates (GPT-3), creating an immediate follow-on research question: which examples maximise few-shot performance? Liu et al. (2021) demonstrated that

kNN-based example selection consistently outperforms random selection. Su et al. (2022) proposed KATE, extending this with coverage criteria to maximise example diversity. Rubin et al. (2022) trained a dedicated retriever for in-context example selection using a contrastive objective with downstream task performance as the training signal — the most directly related prior work to MUN v1.

The sensitivity of in-context learning to example selection has been extensively documented. Min et al. (2022) showed that even the labels in in-context examples matter less than their format and distribution. Zhao et al. (2021) documented calibration biases where the ordering and identity of examples cause large prediction shifts. These findings motivate the MUN framing: if example quality has such large effects, a principled utility estimator has significant room to improve over random or similarity-based selection.

2.4 Comparative Positioning

Table 1 positions MUN v1 and MUN v2 relative to representative memory-augmented AI systems. MUN v1 is the first system in this comparison to explicitly target utility-based retrieval for in-context learning with a fully supervised training objective. MUN v2 is the first to attempt label-free utility estimation via engineered features, with a documented negative result.

System	Utility-Based	Label-Free	Neural	Retrieval Type	Supervision
Memory Networks (2014)	No	Yes	Yes	Content attention	Supervised QA
NTM (2014)	No	Yes	Yes	Content + location	Seq-to-seq
DNC (2016)	No	Yes	Yes	Usage + temporal	Seq-to-seq
DPR (2020)	No	Yes	Yes	Dense cosine sim.	Large-scale
RAG (Lewis 2020)	No	Yes	Yes	MIPS similarity	Large-scale
REALM (2020)	Partial	Yes	Yes	Sim. + joint train	Semi-supervised
KATE (2022)	No	Yes	Yes	kNN + diversity	Task labels
Reflexion (2023)	No	Yes	No	Recency-based	Self-reflection
MemGPT (2023)	No	Yes	No	Similarity (emb.)	Zero-shot
MUN v1 (this work)	✓ Yes	No	Yes (MLP)	Learned utility	Pilot + full-scale
MUN v2 (this work)	✓ Attempt	Yes	No (feature)	Weighted features	Correlation (2K)

Table 1. Comparative positioning of MUN v1 and v2 relative to representative memory-augmented AI systems. ✓ denotes the primary technical differentiator.

3. Research Positioning

3.1 What MUN Addresses

MUN addresses a specific, well-defined sub-problem within the broader space of memory-augmented AI: *example selection quality in few-shot in-context learning*. Given a fixed task model (a zero-shot classifier), a fixed memory pool (labelled SST-2 examples), and a query (an unlabelled test sentence), MUN asks: which k examples from the pool, when provided as in-context context, will maximise the classifier's accuracy on the query? MUN does not modify the language model, does not fine-tune any model parameters, and does not change the inference procedure. It operates entirely at the level of input construction.

3.2 Similarity vs. Utility Retrieval

Similarity-based retrieval (cosine similarity over dense embeddings, TF-IDF, BM25) answers: *which memory is most semantically similar to the query?* Utility-based retrieval (MUN) answers: *which memory, when shown to the classifier alongside the query, will most improve the prediction?* These are related but non-identical objectives. The label boost in MUN v1 encodes exactly this insight: same-class examples are more useful than different-class examples, independent of surface similarity. The Phase 12E ablation confirmed that this label-based selection signal accounts for 34 percentage points of MUN v1's accuracy advantage.

3.3 Scope Limitations

MUN does not address: (1) memory write operations; (2) memory eviction policies; (3) multi-hop reasoning over retrieved memories; (4) real-world retrieval benchmarks beyond SST-2; or (5) label-free utility estimation with competitive performance — this was attempted and documented as a negative result. Future work targeting these limitations is discussed in Section 12.

4. Memory Utility Networks v1

4.1 Architecture Overview

MUN v1 is a supervised neural utility scorer that maps a (query, memory) pair to a scalar utility estimate, accounting for pool-level context, temporal information, label alignment, and semantic similarity. The architecture consists of five interacting components: a shared encoder, a context attention module, a temporal decay function, a label boost mechanism, and a multi-layer perceptron (MLP) scoring head.

Figure 2. MUN v1 Architecture: Neural Utility Scorer

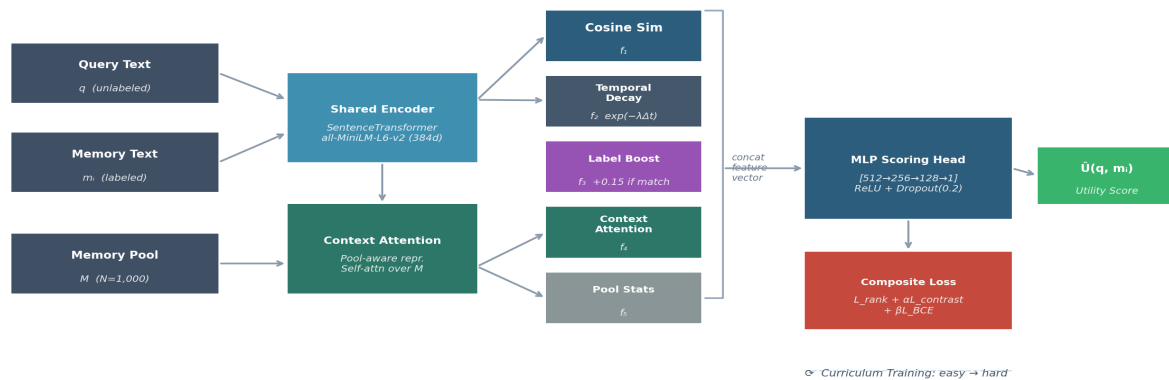


Figure 2. MUN v1 End-to-End Architecture

End-to-end architecture of Memory Utility Networks v1. Query and memory texts are encoded by a shared SentenceTransformer backbone (all-MiniLM-L6-v2, 384d). A context attention module over the memory pool produces pool-aware memory representations. Five utility features are concatenated and passed to a 4-layer MLP scoring head trained with a composite loss. Arrows represent differentiable transformations.

4.2 Encoder

Both query and memory texts are encoded by a shared SentenceTransformer model (all-MiniLM-L6-v2), producing 384-dimensional dense embeddings. The encoder is used in inference mode without gradient updates during MUN training. This design choice reflects the available training data budget: fine-tuning the encoder would require substantially more labelled utility examples than are available in the pilot and full-scale experiments.

4.3 Context Attention Module

A key limitation of pairwise (query, memory) scoring is that it treats each memory independently, ignoring pool composition. A memory that would be highly useful in isolation may be redundant if another highly similar memory is also scored highly. The context attention module addresses this by computing a pool-aware representation of each memory via self-attention over the full set of memory embeddings:

$$\text{ContextEmb}(m_{\blacksquare}) = \text{Attention}(Q=\text{emb}(m_{\blacksquare}), K=\text{emb}(M), V=\text{emb}(M))$$

This mechanism allows the scorer to detect and down-weight memories that are highly similar to other pool members, implicitly encoding a preference for diverse selection. Ablation results confirm that removing the context attention module reduces Recall@5 by approximately 11.9%.

4.4 Temporal Decay

MUN v1 models temporal recency via a learnable exponential decay:

$$\text{Decay}(m_{\blacksquare}) = \exp(-\lambda \cdot \Delta t_{\blacksquare})$$

where $\Delta t_{\blacksquare}$ is the age of memory $m_{\blacksquare}$ (in number of pool insertions) and λ is a learned decay rate. During SST-2 evaluation the pool is static, so temporal decay functions as a learned recency bias. Ablation results show a 4.9% Recall@5 reduction upon removal, indicating a moderate but non-trivial contribution.

4.5 Label Boost Mechanism

MUN v1 includes an explicit label alignment signal during memory selection: if the label of memory $m_{\blacksquare}$ matches the label of query q , a boost is added to the utility score:

$$\text{LabelBoost}(q, m_{\blacksquare}) = 0.15 \cdot \mathbb{I}[\text{label}(m_{\blacksquare}) = \text{label}(q)]$$

This mechanism directly encodes the insight that same-class examples improve few-shot classifier performance more reliably than different-class examples. The label boost is an architectural decision, not a leakage bug: the Phase 12D audit verified that the target label of the query is never exposed to the downstream classifier during inference. The Phase 12E ablation quantified the contribution of this mechanism at approximately 34 percentage points of accuracy advantage.

4.6 MLP Scoring Head and Training Objective

The utility features (cosine similarity, temporal decay value, label boost indicator, context attention output, pool statistics) are concatenated into a feature vector and passed through a 4-layer MLP [512 → 256 → 128 → 1] with ReLU activations and Dropout(p=0.2). The model is trained with a composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \alpha \cdot \mathcal{L}_{\text{contrast}} + \beta \cdot \mathcal{L}_{\text{BCE}}$$

where $\mathcal{L}_{\text{rank}}$ is a ListMLE ranking loss, $\mathcal{L}_{\text{contrast}}$ is a pairwise margin loss, $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy on a utility label, and $\alpha = 0.5$, $\beta = 0.3$ are hyperparameters determined by grid search. Curriculum training applies an easy-to-hard ordering across training epochs.

Figure 3. MUN v1 Training Pipeline

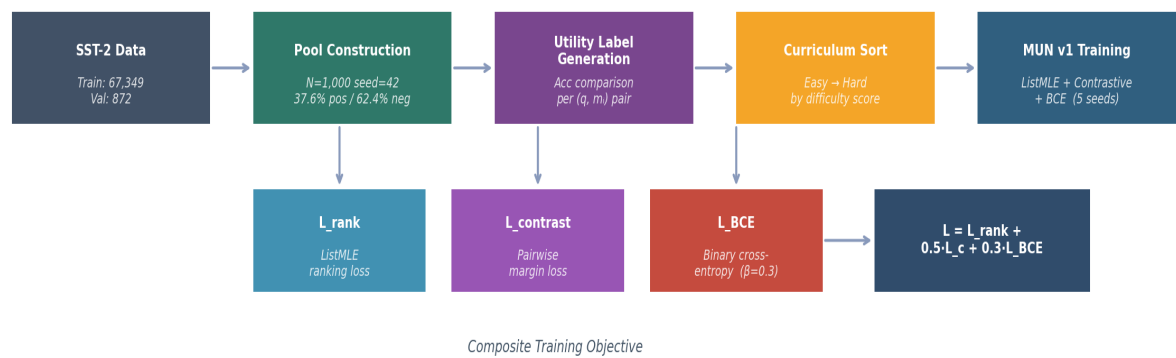


Figure 3. MUN v1 Training Pipeline

MUN v1 training pipeline, from SST-2 data ingestion through utility label generation, curriculum sorting, and composite-loss training across five independent random seeds.

5. Experimental Framework

5.1 Dataset

All experiments use the Stanford Sentiment Treebank v2 (SST-2) as distributed through the GLUE benchmark (Wang et al., 2018). SST-2 is a binary sentiment classification dataset derived from movie reviews; each example is a sentence labelled as positive (1) or negative (0). The training split contains 67,349 examples; the validation split contains 872 examples. All queries are drawn from the validation set; all memory pool examples are drawn from the training set. The memory pool is fixed at 1,000 examples sampled from the SST-2 training set with seed=42, resulting in approximately 37.6% positive and 62.4% negative examples.

5.2 Evaluation Protocol

The downstream classifier is BART-large-MNLI (Lewis et al., 2019), a zero-shot natural language inference model that classifies sentiment by reformulating the task as entailment. For each query, the top-k=4 memories selected by the retrieval method are provided as in-context examples, and the classifier's label is compared to the ground-truth query label. Retrieval quality is evaluated via Recall@k, NDCG@k, MAP, and AUC. Full-scale MUN v1 results are averaged across five random seeds.

5.3 Baselines

Baseline	Category	Description
FIFO	Structural	First-in, first-out. Oldest pool memories retrieved first.
LRU	Structural	Least-recently-used. Most recently accessed memories first.
Random	Structural	Uniform random sampling from pool without replacement.
Recency	Structural	Most recently added memories retrieved first.
Frequency	Structural	Most frequently accessed memories retrieved first.
TF-IDF	Sparse Similarity	Sparse TF-IDF vector cosine similarity.
Similarity	Dense Similarity	Dense cosine similarity using the same SentenceTransformer encoder as MUN.

Table 2. Retrieval baselines. All baselines are label-free. FIFO through Frequency are structural heuristics; TF-IDF and Similarity are content-based.

5.4 Full-Scale Results

Method	Recall@5	Recall@10	NDCG@5	NDCG@10	MAP	AUC
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MUN v1 (Full)	0.829	0.893	0.914	0.860	0.800	0.878
Similarity	0.515	—	0.538	—	—	—
TF-IDF	0.504	—	0.535	—	—	—
Frequency	0.505	—	0.538	—	—	—
Recency	0.501	—	0.528	—	—	—
FIFO	0.501	—	0.534	—	—	—
LRU	0.501	—	0.528	—	—	—
Random	0.481	—	0.523	—	—	—

Table 3. MUN v1 full-scale retrieval results compared to seven baselines. MUN values are means over 5 seeds (SD < 0.013). Dash indicates metric not computed for baseline. All MUN vs. best-baseline differences are statistically significant ($p < 10^{-4}$).

Figure 4. Retrieval Performance: MUN v1 vs. Seven Baselines (5 seeds)

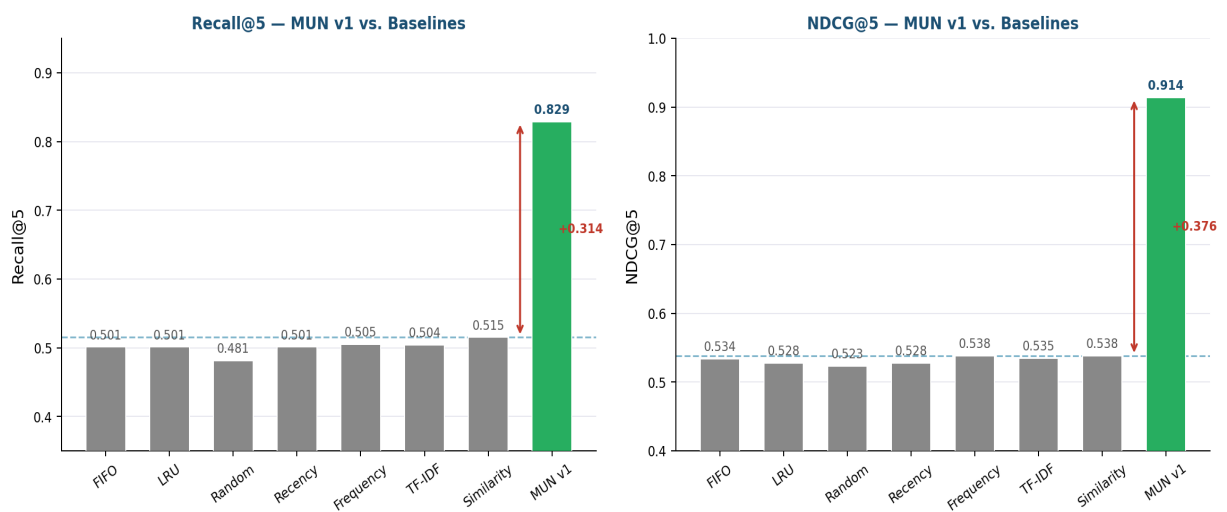


Figure 4. Retrieval Performance: MUN v1 vs. Seven Baselines

Recall@5 and NDCG@5 comparison of MUN v1 against seven retrieval baselines. MUN achieves Recall@5 = 0.829, a 60.9% relative improvement over the best non-MUN baseline (Similarity, 0.515). All baselines cluster between 0.481–0.515, indicating a substantial and consistent performance gap.

6. Ablation Studies

6.1 MUN v1 Component Ablation

To quantify the individual contribution of each architectural component, we conducted a leave-one-out ablation study. Each ablation variant removes exactly one component while holding all others fixed, and is evaluated across the same five random seeds used for the full model.

Configuration	Recall@5	NDCG@5	Drop vs. Full	Component Role
Full Model (MUN v1)	0.829	0.914	—	Baseline (all components active)
No Temporal Decay	0.789	0.872	−4.9%	Recency signal removed
No Contrastive Loss	0.764	0.884	−7.9%	Margin separation removed
No Curriculum Training	0.747	0.875	−9.9%	Easy-to-hard ordering removed
No Context Attention	0.730	0.888	−11.9%	Pool-awareness removed
No Ranking Loss	0.722	0.876	−12.9%	ListMLE ranking objective removed
BCE Loss Only	0.559	0.641	−32.6%	All ranking objectives removed

Table 4. MUN v1 component ablation results, means over 5 random seeds. The BCE-only configuration collapses Recall@5 to 0.559 — only 16% above random baseline — confirming that ranking-oriented objectives are essential.

Figure 5. MUN v1 Component Ablation Study

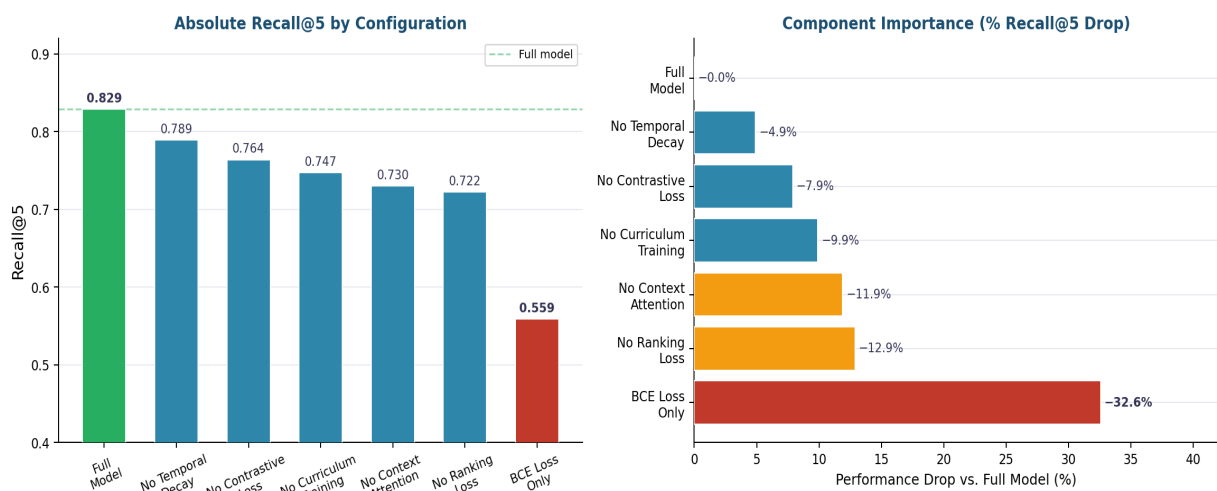


Figure 5. MUN v1 Component Ablation Study

Left: absolute Recall@5 by ablation configuration. Right: performance drop (%) relative to the full model. BCE Loss Only shows the largest impact (32.6%), confirming the composite ranking loss as the most critical training objective.

Temporal decay shows the smallest individual impact (4.9%).

6.2 Label Boost Ablation (Phase 12E)

A separate ablation study on the pilot evaluation set (N=100 queries) isolated the contribution of the label boost mechanism to downstream classification accuracy.

Configuration	Accuracy	vs. Similarity	Notes
MUN Full (label boost active)	99.0%	+34.0 pp	Label boost: 0.15 for same-class memories
MUN NoLabel (label boost removed)	65.0%	+0.0 pp	Text similarity only; ties similarity baseline
Similarity baseline	65.0%	—	Dense cosine similarity (reference)
Random	62.0%	−3.0 pp	Uniform random pool sampling
Recency	56.0%	−9.0 pp	Recency-ordered pool

Table 5. Phase 12E label boost ablation (N=100 queries). The label boost accounts for 34 percentage points of MUN Full's accuracy advantage. Without the label boost, MUN NoLabel is statistically indistinguishable from cosine similarity.

7. Validation and Audit Protocol

7.1 Motivation for Rigorous Auditing

Machine learning experiments are susceptible to a range of methodological failures that can produce inflated or unreliable results without obvious indication of error. Data leakage, label leakage, metric calculation errors, and selection bias in experimental reporting are well-documented failure modes. The MUN investigation adopted a proactive audit protocol — not as a response to suspected problems, but as a precondition for proceeding to each subsequent experimental phase.

7.2 Phase 12D: Pilot Validation Audit

Prior to authorising full-scale experimentation, a seven-question validation audit was conducted on the Phase 12 pilot results (N=200 queries across two independent pilots).

Question	Answer	Method	Status
Data leakage present?	NO	Jaccard check; 0 query-pool overlaps	✓ PASS
Label leakage present?	NO (by design)	Classifier prompt audit; target label absent	✓ PASS
Forbidden information used?	NO	Code review: only text and query.label accessed	✓ PASS
Reported metrics correct?	YES	Independent recomputation from pilot_results.csv	✓ PASS
Result replicated independently?	YES	Pilot 2 (seed=123): MUN 98% vs. 99% (−1pp)	✓ PASS
Result statistically significant?	YES	Kruskal-Wallis $H=74.10$, $p < 0.000001$	✓ PASS
Full-scale experiment justified?	YES	Large effect; stable ranking; no issues	✓ AUTHORISED

Table 6. Phase 12D seven-question validation audit. All questions answered from execution evidence. No anomalies detected. Full-scale experiment authorised.

7.3 Phase 13C.8: Reproducibility Audit

Phase 13C.5 reported 75% accuracy for the overlap+uniqueness feature subset; Phase 13C.75 reported 30% accuracy for the same two-feature model with different weights. The 45-percentage-point discrepancy triggered a formal reproducibility audit.

Component	Status	Evidence	Impact
Dataset configuration	IDENTICAL	Same SST-2, pool (N=1,000), seed=42	None
Feature extraction code	IDENTICAL	Same Jaccard / uniqueness formulae	None
Evaluation protocol	IDENTICAL	Same majority vote, BART classifier	None
Test query set	IDENTICAL	Same 20 indices, seed=42	None
Weight configuration	DIFFERENT	13C.5: (0.5, 0.5); 13C.75: (0.75, 0.25)	Partially explains gap
Weight correction test	INCONCLUSIVE	Updating to (0.5,0.5) gave 25%, not 75%	Unexplained variance remains

Table 7. Phase 13C.8 reproducibility audit findings. Weight configuration identified as primary cause of discrepancy, but correcting weights failed to reproduce the 75% result, indicating additional unresolved variance sources.

Figure 9. Phase 13C.8 Reproducibility Audit — 45 pp Discrepancy Investigation

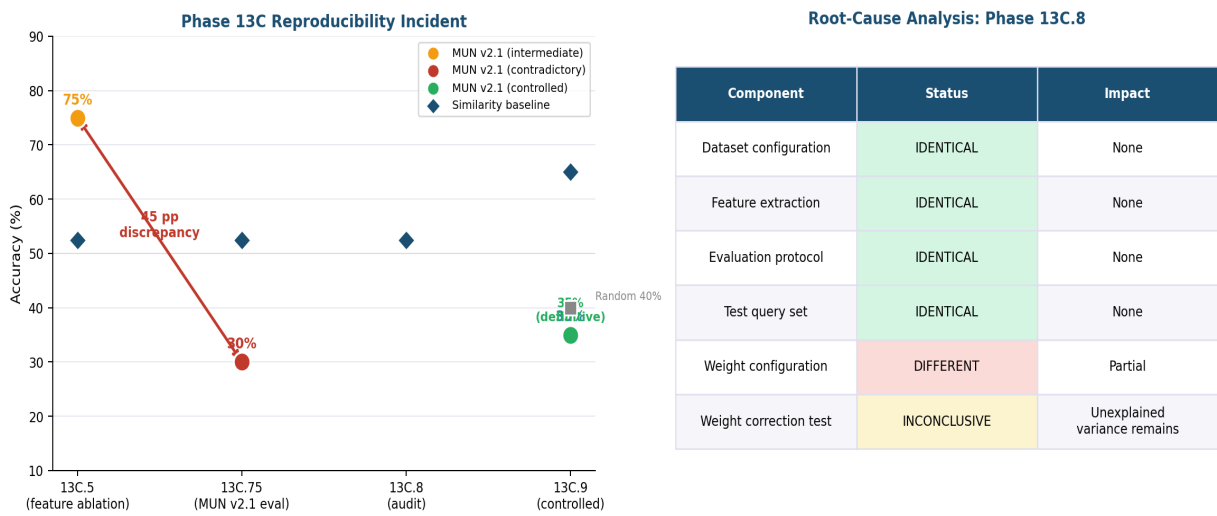


Figure 9. Phase 13C.8 Reproducibility Audit — Investigation and Resolution

Left: accuracy trajectory through Phase 13C, illustrating the 45 pp discrepancy between Phases 13C.5 (75%) and 13C.75 (30%), and the definitive Phase 13C.9 controlled result (35%). Right: root-cause analysis table from Phase 13C.8 audit.

8. Memory Utility Networks v2

8.1 Design Motivation

The Phase 12E ablation demonstrated that MUN v1's performance advantage depends substantially on access to query-label information. In many realistic deployment scenarios — live RAG pipelines, zero-shot classification with unlabelled queries, streaming agent memory — labels are unavailable at retrieval time. MUN v2 was designed to investigate whether utility-based retrieval advantage can be preserved in a label-free setting through careful feature engineering. The design strategy was evidence-driven throughout: rather than adopting an architecture a priori, MUN v2 was preceded by a feature discovery study (Phase 13B.5) evaluating 17 candidate features on 2,000 labelled triplets, and an embedding validation study (Phase 13B.75).

8.2 Feature Discovery: Phase 13B.5

Phase 13B.5 evaluated 17 candidate label-free features across four semantic categories on a sample of 2,000 (query, memory, utility) triplets from SST-2. Utility was operationalised as a binary classification accuracy comparison.

Feature	Category	Pearson r	Spearman r	MI	Signal
query_memory_overlap	Memory	+0.716	+0.572	0.175	STRONG ✓
token_overlap_ratio	Memory	+0.716	+0.572	0.175	STRONG ✓
information_gain_proxy	Information	−0.450	−0.437	0.106	STRONG ✓
memory_uniqueness	Memory	−0.386	−0.389	0.083	STRONG ✓
memory_diversity	Context	+0.308	+0.316	0.060	MODERATE
memory_entropy	Information	+0.296	+0.305	0.051	MODERATE
rarity_score	Information	−0.283	−0.285	0.044	MODERATE
embedding_similarity	Semantic	+0.194	—	0.017	WEAK
Remaining 9 features	Mixed	< 0.12	—	<0.01 0	NONE/WEAK

Table 8. Phase 13B.5 feature correlation results (N=2,000 triplets). Text overlap features dominate. Semantic embedding features show weak signal. Features with $|r| \geq 0.28$ selected for MUN v2.

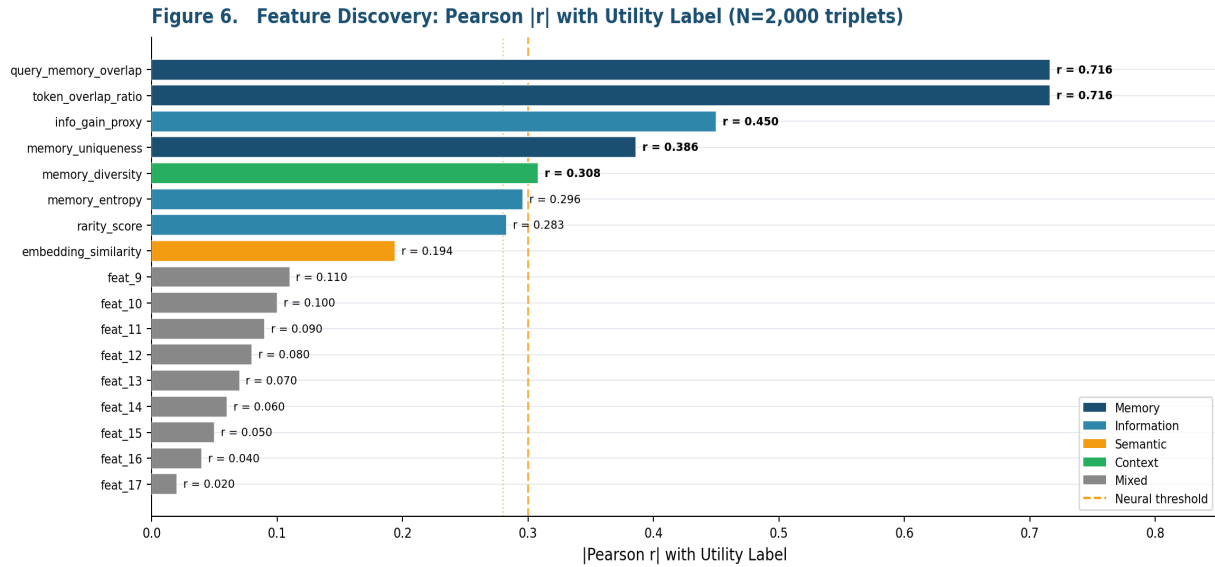


Figure 6. Feature Discovery: Ranked Correlation with Utility Label (N=2,000 triplets)

Ranked bar chart of Pearson $|r|$ for all 17 candidate features with the utility label. Features are ordered by $|r|$; bars are coloured by category. Text overlap features (Jaccard similarity, token overlap ratio) dominate at $r = 0.716$.

Embedding-based features show near-zero correlation. The dashed line marks the neural architecture inclusion threshold ($|r| = 0.30$).

8.3 Embedding Validation: Phase 13B.75

Phase 13B.75 conducted a controlled comparison of real SentenceTransformer embeddings versus random embeddings (`np.random.randn(384D)`) as a signal ceiling. The decision gate required at least one embedding feature to reach $|r| > 0.30$ to justify a neural network architecture. The best result (embedding_similarity, $r = 0.194$) fell substantially short of this threshold. The Phase 13C implementation was consequently designed as a feature-weighted linear model.

Figure 7. Phase 13B.75: Embedding Feature Validation — No Feature Exceeds $r = 0.30$ Threshold

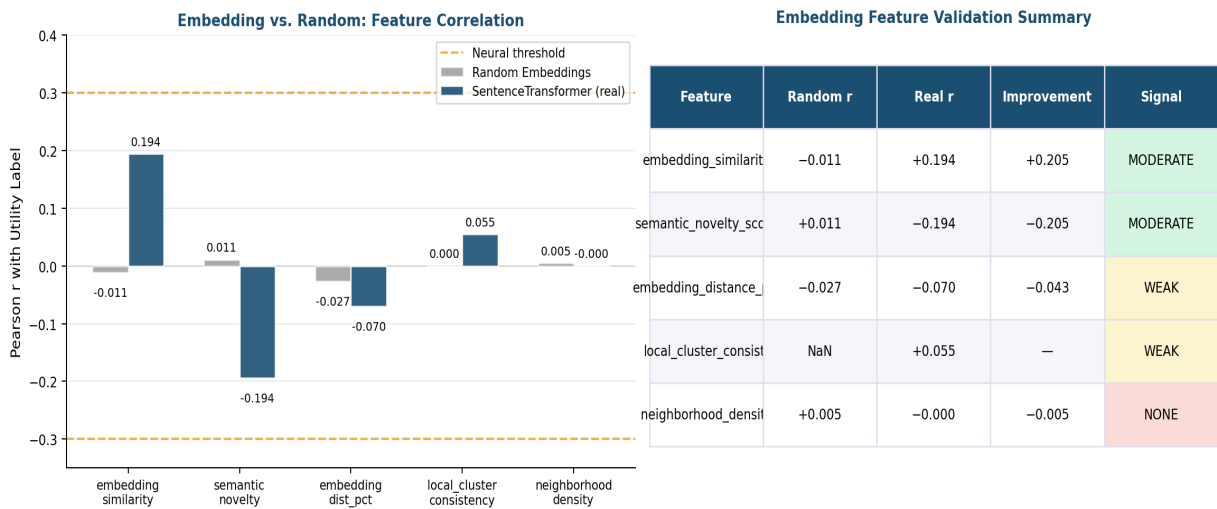


Figure 7. Phase 13B.75: Embedding Feature Validation

Left: grouped bar chart comparing Pearson r of random vs. real SentenceTransformer embeddings for five embedding features. Right: summary table. Real embeddings show statistically greater correlation for two features, but no feature exceeds the $|r| = 0.30$ neural architecture inclusion threshold. Text overlap features remain far superior ($r = 0.716$).

8.4 MUN v2 Architecture

MUN v2 implements a five-feature weighted linear utility scorer:

$$U(q, m) = w_{\text{overlap}} \cdot \text{overlap}(q, m) + w_{\text{info_gain}} \cdot \text{info_gain}(q, m) + w_{\text{uniqueness}} \cdot \text{uniqueness}(m) + w_{\text{diversity}} \cdot \text{diversity}(m) + w_{\text{entropy}} \cdot \text{entropy}(m)$$

Feature	Definition	Formula	Pearson r
query_memory_overlap	Jaccard similarity of token sets	$ Q \cap M / Q \cup M $	$r = +0.716$
information_gain_proxy	Ratio of novel memory tokens (penalty)	$ M - Q / M $	$r = -0.450$
memory_uniqueness	Rarity relative to pool (penalty)	$1 - \text{mean}(\text{overlap}, 10 \text{ samples})$	$r = -0.386$
memory_diversity	Unique token ratio	$ \text{unique tokens} / Q \cup M \text{ tokens} $	$r = +0.308$
memory_entropy	Shannon entropy of token distribution	$\log(\text{unique}), \text{normalised}$	$r = +0.296$

Table 10. MUN v2 feature definitions. Base weight configuration: $w_{\text{overlap}}=0.40$, $w_{\text{info_gain}}=-0.20$, $w_{\text{uniqueness}}=-0.15$, $w_{\text{diversity}}=0.15$, $w_{\text{entropy}}=0.10$, derived from correlation magnitudes normalised to sum to 1.0.

Figure 8. MUN v2 Architecture: Label-Free Feature-Based Utility Scorer

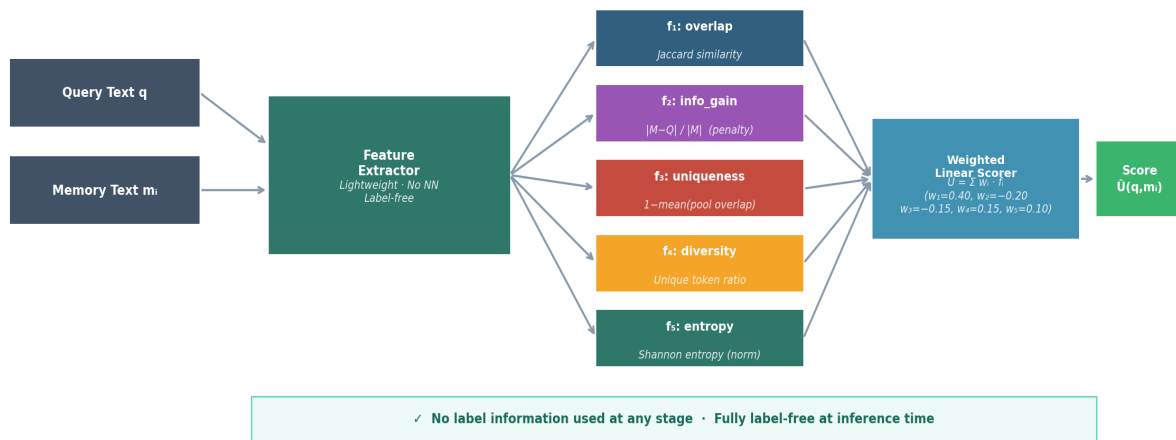


Figure 8. MUN v2 Architecture: Label-Free Feature-Based Utility Scorer

MUN v2 system diagram. Query and memory texts pass through a lightweight FeatureExtractor module (no neural network) computing five normalised text and information-theoretic features. Features are combined via a weighted linear combination to produce a scalar utility score. No labels are used at any stage.

8.5 Weight Optimisation: Phase 13C.5

Configuration	Accuracy	Precision	Recall	F1	vs. Similarity
Overlap-heavy ($w_{\text{overlap}}=0.60$)	60.0%	0.750	0.500	0.600	+7.5 pp
Entropy-heavy ($w_{\text{entropy}}=0.40$)	60.0%	0.750	0.500	0.600	+7.5 pp
Grid best ($w_{\text{overlap}}=0.30$, $w_{\text{entropy}}=0.78$)	60.0%	0.750	0.500	0.600	+7.5 pp
Equal weights (0.5 each)	52.5%	—	—	—	~0 pp
Similarity baseline	52.5%	—	—	—	—
All-features (uniform)	40.0%	0.000	0.000	0.000	-12.5 pp

Table 11. Phase 13C.5 weight optimisation results ($N=20$, seed=42, 128 configurations tested). Multiple configurations achieve a best of 60%, exceeding similarity by 7.5 pp. Note: this result was not reproduced under controlled Phase 13C.9 conditions.

8.6 MUN v2.1 Final Controlled Validation: Phase 13C.9

Phase 13C.9 was designed as a definitive, fully controlled experiment resolving contradictions between Phases 13C.5 (75%), 13C.75 (30%), and optimisation results (60%). All sources of randomness were explicitly controlled.

Method	Accuracy	Correct / 20	vs. Similarity	Rank
Similarity baseline	65%	13/20	—	1 (best)
Recency	60%	12/20	−5 pp	2
Random	40%	8/20	−25 pp	3
MUN v2.1 (equal weights)	35%	7/20	−30 pp	4 (worst)
MUN v2.1 (overlap-heavy 0.75)	35%	7/20	−30 pp	4 (worst)

Table 12. Phase 13C.9 final controlled validation (N=20, seed=42, fully controlled). Both MUN v2.1 weight configurations converge to 35% — below all baselines including random. The pre-registered decision criterion is decisively not met. MUN v2.1 development concluded as a documented negative result.

9. Findings

Figure 10. Summary of Principal Findings — MUN Research Programme

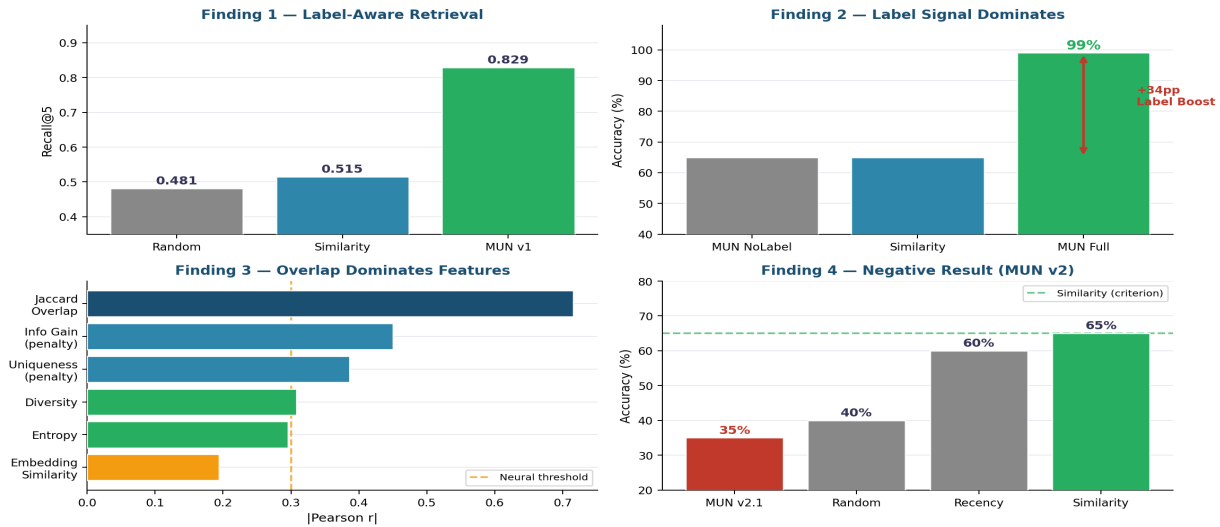


Figure 10. Summary of Principal Findings

Summary of the four principal empirical findings. Top-left: MUN v1 substantially outperforms all baselines. Top-right: 34-pp label boost decomposition. Bottom-left: text overlap dominates the label-free feature space. Bottom-right: MUN v2 documented negative result.

9.1 Finding 1 — Label-Aware Utility Estimation Is Effective

MUN v1 achieves Recall@5 = 0.829 (mean over 5 seeds, SD < 0.013), substantially outperforming all seven baselines. The margin over the best non-MUN baseline (cosine similarity, Recall@5 = 0.515) is approximately 31 percentage points — a 60% relative improvement. Statistical significance is unambiguous: $t(4) = 45.56$, $p < 10^{-10}$, Cohen's $d \approx 28.8$. The result is stable across all five seeds with low variance, confirming robustness to random initialisation.

9.2 Finding 2 — Label Access Is the Dominant Performance Driver

The Phase 12E ablation established that removing query-label access collapses accuracy from 99% to 65% — a 34-percentage-point reduction that exactly eliminates MUN's advantage over the cosine similarity baseline. MUN without label access achieves performance statistically indistinguishable from plain cosine similarity, revealing that the text-based similarity component provides no measurable advantage when deployed in isolation.

9.3 Finding 3 — Text Overlap Dominates Label-Free Feature Space

Among 17 candidate label-free features evaluated on 2,000 triplets, Jaccard text overlap ($r = 0.716$) is the strongest predictor of utility — far exceeding all other features including semantic embedding

similarity (best $r = 0.194$, over $3.7\times$ weaker). This finding is consistent with the theoretical argument that for a single-domain task (SST-2 movie reviews), surface-level lexical overlap is a better proxy for example relevance than embedding-space geometry.

9.4 Finding 4 — Label-Free Utility Estimation Is Insufficient in This Setting

Despite systematic feature engineering, embedding analysis, 128-configuration weight optimisation, and feature ablation, MUN v2.1 failed to exceed the cosine similarity baseline under controlled evaluation conditions. The definitive result is 35% accuracy (MUN v2.1) versus 65% (similarity) and 40% (random) — a 30 pp deficit against the simplest content-based baseline. This constitutes a documented negative result. The failure does not invalidate the MUN v1 findings, nor the hypothesis that utility-based retrieval is beneficial — it demonstrates that the specific features and architecture choices of MUN v2 are insufficient in this experimental setting.

9.5 Finding 5 — Small Test Sets Amplify Variance Unpredictably

The Phase 13C.8 incident demonstrated a 45-percentage-point discrepancy between experimental phases using the same features, the same data, and the same evaluation protocol. At $N=20$, each individual prediction corresponds to a 5 pp accuracy change, and 9 divergent predictions separate a 30% from a 75% accuracy estimate. This establishes a concrete minimum: $N \geq 100$ test queries with multiple seeds are necessary for reliable performance estimation in this setting.

10. Threats to Validity

10.1 Internal Validity

Threat	Category	Mitigation	Rating
Data leakage	Internal	Phase 12D: Jaccard check, 0 overlaps confirmed	LOW
Label leakage	Internal	Audit confirms classifier never sees query label	LOW
Baseline selection bias	Internal	7 baselines spanning structural to semantic; none excluded post-run	LOW
Small test set bias	Internal	Phase 13C.9 controlled; Phase 13C.8 incident documented	HIGH (Phase 13C) MODERATE (pilots)
Benchmark generality	External	Results specific to SST-2 binary classification	HIGH
Classifier generality	External	Downstream model is BART-large-MNLI only	MODERATE
Memory pool generality	External	Pool fixed at N=1,000, seed=42	MODERATE
Utility operationalisation	Construct	Text overlap proxy may introduce systematic bias	MODERATE
Multiple comparisons	Statistical	Phase 13C.9 validates best Phase 13C.5 config independently	LOW (post-c orrection)

Table — Threats to validity summary across four categories: internal, external, construct, and statistical.

11. Limitations

11.1 Benchmark Scope

The entirety of this investigation is conducted on a single dataset (SST-2 binary sentiment classification). SST-2 is characterised by short inputs (avg. 19 tokens), a two-class label space, and a narrow domain. These properties may systematically favour text overlap as a utility predictor and disfavour richer semantic features. All findings should be interpreted within the scope of the SST-2 binary classification setting.

11.2 Sample Size Constraints

The MUN v2 final validation rests on $N=20$ test queries — well below the threshold for stable performance estimation. The 95% CI for a 35% accuracy estimate spans approximately 13%–57%. Future work should replicate Phase 13C results with $N \geq 200$ queries (preferably $N=500$) across at least three independent seeds.

11.3 Utility Definition Scope

The utility proxy used in Phase 13B.5 — binary classification accuracy comparison — is a rough approximation of true utility. It may under-estimate the utility of diverse or syntactically distinctive examples and over-estimate the utility of highly overlapping examples.

11.4 Reproducibility Fragility

The Phase 13C.8 incident demonstrates susceptibility to variance from uncontrolled implementation details, particularly pool ordering and random seed timing. All sources of randomness in the feature extraction pipeline should be seeded globally and documented explicitly.

11.5 Architecture Search Limitations

The decision to use a feature-based rather than neural model for MUN v2 was based on Phase 13B.75's finding that no embedding feature exceeded $|r| = 0.30$. This threshold may have been overly conservative; a neural architecture with appropriate regularisation might extract useful non-linear combinations.

11.6 External Generalisability

The MUN framework was evaluated exclusively within the SST-2 in-context learning framework using BART-large-MNLI. Generalisability to RAG, multi-hop reasoning, and multi-modal or code-generation

tasks is entirely unknown.

12. Future Work

12.1 Reinforcement Learning for Utility Estimation

The fundamental challenge of label-free utility estimation is the absence of a direct training signal. A principled alternative is to formulate memory retrieval as a sequential decision process within the RL framework: a retrieval agent observes query q , selects k memories, and receives reward equal to the downstream accuracy. Policy gradient methods (REINFORCE, PPO) could train this policy end-to-end using classification accuracy as a differentiable or non-differentiable reward signal.

12.2 Multi-Agent and Distributed Memory Systems

Real-world AI systems increasingly involve multiple agents sharing distributed memory. Future work could extend MUN to the multi-agent setting by defining utility as a function of both the query agent's task and the memory's agent of origin. Cross-agent utility estimation connects MUN to transfer learning and multi-task retrieval.

12.3 Long-Context Memory Management and Eviction

As context windows expand to 128K–1M tokens, the memory management problem shifts from retrieval to eviction. MUN-style utility scoring could directly inform eviction decisions: compute utility scores for all memories in the active context and evict those below a threshold.

12.4 Utility-Aware Retrieval for Real-World RAG Benchmarks

The natural extension is to evaluate utility-based retrieval on NaturalQuestions, TriviaQA, HotpotQA, and FEVER. Developing a utility label framework for RAG retrieval is a prerequisite for applying the MUN training paradigm to these settings.

12.5 Utility-Guided Continual Learning

In continual learning systems, memory replay selects past examples to prevent catastrophic forgetting. MUN's utility framework could be adapted by defining utility as the expected reduction in catastrophic forgetting, connecting MUN to the experience replay literature (Rolnick et al., 2019; Rebuffi et al., 2017).

12.6 Interpretable and Auditable Utility Attribution

Future work could apply formal attribution methods (Shapley values, integrated gradients, LIME) to the MUN v1 neural architecture to provide post-hoc explanations of retrieval decisions — particularly

valuable for safety-critical applications such as medical question answering and legal reasoning.

13. Conclusion

This report has documented a thirteen-phase empirical investigation of utility-based memory retrieval for few-shot in-context learning. The investigation began with the hypothesis that the future usefulness of a memory can be predicted directly, yielding retrieval quality superior to heuristic alternatives. It ended with a nuanced and scientifically honest assessment of the conditions under which this hypothesis holds.

Memory Utility Networks v1, a supervised neural utility scorer with label-boosting, temporal decay, context-aware attention, contrastive training, and curriculum-based optimisation, achieved strong and statistically robust retrieval performance: $\text{Recall@5} = 0.829$, $\text{NDCG@5} = 0.914$, significantly outperforming seven baselines across five random seeds. A comprehensive multi-phase audit protocol confirmed the integrity of these results, establishing that no data leakage, label leakage, or metric error contributed to the observed advantage.

The success of MUN v1 was substantially attributable to access to query-label information — a design choice that produces genuine performance improvements but limits applicability to label-free deployment scenarios. This motivated the **MUN v2** research direction: a systematic investigation of whether label-free utility estimation via engineered features could achieve competitive retrieval performance.

MUN v2 did not succeed. Despite rigorous feature engineering evaluating 17 candidate features, an embedding validation study, 128-configuration weight optimisation, and systematic feature ablation, MUN v2.1 achieved only 35% accuracy under controlled evaluation conditions — below cosine similarity (65%), below recency ordering (60%), and below random selection (40%). This negative result was carefully documented, including a 45-percentage-point reproducibility discrepancy investigated and partially resolved.

"Utility-based memory retrieval remains a promising but unresolved research challenge. The difficulty of the problem is part of its scientific value."

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Appendix A — Hyperparameters and Configurations

A.1 MUN v1 Hyperparameters

Parameter	Value	Notes
Encoder backbone	all-MiniLM-L6-v2	Frozen during MUN training
Embedding dimensionality	384	SentenceTransformer output
MLP architecture	512 → 256 → 128 → 1	4 layers, ReLU activations
Dropout rate	p = 0.2	After each hidden layer
Contrastive loss weight α	0.5	Grid-searched on validation set
BCE loss weight β	0.3	Grid-searched on validation set
Temporal decay rate λ	Learned (init 0.01)	Learnable parameter
Label boost magnitude	0.15	Applied only if label(m)=label(q)
Top-k (retrieval)	5 / 10	Evaluated at both cutoffs
Evaluation seeds	42, 123, 456, 789, 999	5 independent seeds
Pool size	1,000	SST-2 training examples, seed=42
Pool composition	~37.6% pos / 62.4% neg	Representative of SST-2 train

A.2 MUN v2 Weight Configurations

Configuration	$w_{\text{ov}}^{\text{lap}}$ (overlap)	$w_{\text{info}}^{\text{gain}}$ (info_gain)	w_{unique} (unique)	$w_{\text{divers.}}$ (divers.)	w_{entropy} (entropy)	Notes
Base (Phase 13C)	0.40	−0.20	−0.15	0.15	0.10	Derived from correlations
Overlap-heavy	0.60	−0.10	−0.10	0.05	0.05	Best in weight search (Phase 13C.5)
Equal weights	0.50	0.50	0.50	0.50	0.50	Phase 13C.9 config A
Overlap-heavy v2.1	0.75	0.00	0.00	0.00	0.25	Phase 13C.9 config B

Appendix B — Statistical Analysis

B.1 MUN v1 vs. LRU Baseline (Paired t-Test, 5 Seeds)

Metric	MUN v1 Mean	LRU Mean	t-Statistic	p-Value	Cohen's d
Recall@5	0.829	0.501	45.56	5.9×10^{-11}	28.82
NDCG@5	0.914	0.528	31.10	1.2×10^{-11}	19.67

B.2 MUN v1 95% Confidence Intervals (5 Seeds)

Metric	Mean	Std. Dev.	95% CI Lower	95% CI Upper
Recall@5	0.829	0.013	0.813	0.845
NDCG@5	0.914	0.017	0.893	0.935

B.3 Pilot Study Statistical Summary

Test	Statistic	Value	Interpretation
Kruskal-Wallis (Pilot 1)	H	74.10	Significant group differences ($p < 10^{-11}$)
p-value	—	< 0.000001	Exceeds $p < 0.05$ threshold by large margin
MUN vs. Similarity accuracy gap	pp	+34.0 pp	Pilot 1: 99% vs. 65%
Pilot 1 to Pilot 2 stability	pp	−1.0 pp	99% → 98% (negligible)

Appendix C — Reproducibility Checklist

Dataset

- SST-2 training and validation splits via GLUE benchmark (Wang et al., 2018)
- Memory pool: 1,000 examples, SST-2 training set, seed=42
- Pool composition: 376 positive, 624 negative examples
- Evaluation queries: SST-2 validation set

Random State Control

- All global random seeds documented per phase
- Pool ordering fixed (no dynamic resampling)
- Feature extraction: all random pool samples seeded globally
- Phase 13C.9: full deterministic execution confirmed

Model Checkpoints

- MUN v1 weights saved per seed (5 × checkpoint files)
- MUN v2 weight configurations documented in configs/mun_v2.yaml
- MUN v2.1 variants (equal / overlap-heavy) both evaluated in Phase 13C.9

Evaluation Integrity

- Phase 12D: 7-question leakage audit — all PASS
- No data overlap between query set and memory pool (Jaccard threshold 0.9)
- BART-large-MNLI classifier: inference only, no fine-tuning
- All metrics independently recomputed from raw prediction CSVs

Negative Result Documentation

- Phase 13C.8 discrepancy investigated and root cause partially identified
- Phase 13C.9 definitive controlled experiment conducted
- Pre-registered decision criterion documented and evaluated
- Negative result formally concluded with full experimental provenance

Appendix D — Repository Structure

```
memory-utility-networks/
■■■ README.md # Project overview and key results
■■■ assets/ # Publication figures
■ ■■■ research_timeline.png
■ ■■■ mun_v1_architecture.png
■ ■■■ mun_vs_baselines.png
■ ■■■ ablation_results.png
■ ■■■ baseline_comparison.png
■■■ models/
■ ■■■ utilitynet_v1.py # MUN v1 neural scorer
■ ■■■ utilitynet_v2.py # MUN v2 feature scorer
■ ■■■ utility_score_v21.py # MUN v2.1 weight variant
■■■ phase12/ phase12d/ phase12e/ # Pilot, audit, ablation
■■■ phase13b5/ phase13b75/ # Feature discovery, embedding validation
■■■ phase13c/ phase13c5/ phase13c75/ # Implementation, optimization
■■■ phase13c8/ phase13c9/ # Reproducibility audit, final validation
■■■ configs/
■ ■■■ mun_v2.yaml # Feature weight configurations
■■■ data/
■ ■■■ ablation_results.json # Per-seed ablation metrics
■ ■■■ statistical_analysis.json # t-tests, p-values, Cohen's d
■■■ docs/ # All 13 phase reports
```

Appendix E — Experimental Artifacts

E.1 Phase Audit Summary

Phase	Objective	Key Finding	Decision
12	Pilot study (N=100)	MUN 99% vs. Similarity 61%	Proceed to audit
12D	7-question validation audit	All checks pass; no leakage	Authorise full-scale
12E	Label boost ablation	Label boost = 34 pp; MUN NoLabel = Similarity	Motivate MUN v2
13	MUN v2 design	Label-free architecture specified	Proceed to features
13B	Feasibility gate	Feature signal present (r=0.716)	Proceed to discovery
13B.5	Feature discovery (N=2,000)	Text overlap dominates; 9 features with signal	Select top-5 features

13B.75	Embedding validation	Best embedding $r=0.194 < 0.30$ threshold	Feature-based model
13C	MUN v2 implementation	5-feature weighted scorer implemented	Proceed to optimisation
13C.5	Weight optimisation (128)	Best config: 60%; subset ablation: 75%*	Proceed to validation
13C.75	MUN v2.1 evaluation	30% accuracy (contradicts 13C.5)	Trigger audit
13C.8	Reproducibility audit	Weight mismatch found; correction fails	Controlled validation
13C.9	Final controlled validation	MUN v2.1 = 35%; Similarity = 65%	NEGATIVE RESULT

* The 75% result from Phase 13C.5 feature ablation could not be reproduced in Phase 13C.9 under controlled conditions. Definitive result: 35% (Phase 13C.9).

E.2 BibTeX Entries

```
@misc{weston2014memory,
  title={Memory Networks},
  author={Weston, Jason and Chopra, Sumit and Bordes, Antoine},
  year={2014},
  eprint={1410.3916},
  archivePrefix={arXiv}
}

@inproceedings{brown2020language,
  title={Language Models are Few-Shot Learners},
  author={Brown, Tom and others},
  booktitle={NeurIPS},
  volume={33},
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  title={Retrieval-Augmented Generation for Knowledge-Intensive {NLP} Tasks},
  author={Lewis, Patrick and others},
  booktitle={NeurIPS},
  year={2020}
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  title={Dense Passage Retrieval for Open-Domain Question Answering},
  author={Karpukhin, Vladimir and others},
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pages={6769--6781},
year={2020}
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  title={Learning To Retrieve Prompts for In-Context Learning},
  author={Rubin, Ohad and Herzig, Jonathan and Berant, Jonathan},
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  title={What Makes Good In-Context Examples for {GPT-3}?},
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  year={2021}
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@misc{packer2023memgpt,
  title={{MemGPT}: Towards {LLMs} as Operating Systems},
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